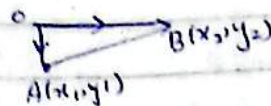


VECTORS

If $A(x_1, y_1)$ & $B(x_2, y_2)$
 $\vec{AB} = (x_2 - x_1)\vec{i} + (y_2 - y_1)\vec{j}$

Also,

$$\vec{AB} = \vec{OB} - \vec{OA}$$



$\vec{r}, \bar{r}, \underline{r}, r$

$$\vec{OA} + \vec{AB} = \vec{OB}$$

$$\vec{AB} = \vec{OB} - \vec{OA}$$

Magnitude / Norm / length

$$\vec{OP} = \vec{r} = 3\vec{i} + 4\vec{j}$$

$$|\vec{r}| = r = \sqrt{x^2 + y^2} = \sqrt{3^2 + 4^2} = 5$$

Unit vectors: A vector of magnitude 1 is called unit vector.

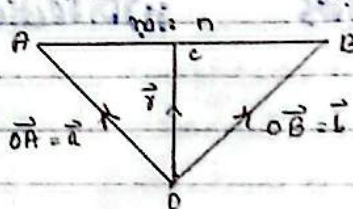
It is defined as $\hat{r} = \frac{\vec{r}}{|\vec{r}|}$

Ratio Formula:

$$\vec{r} = \frac{m\vec{b} + n\vec{a}}{m+n}$$

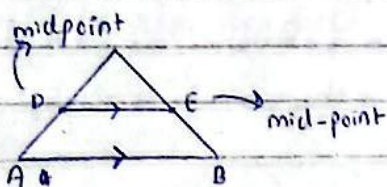
If C is the mid point of $\vec{AB}$

$$\therefore \vec{r} = \frac{\vec{a} + \vec{b}}{2}$$



Note:

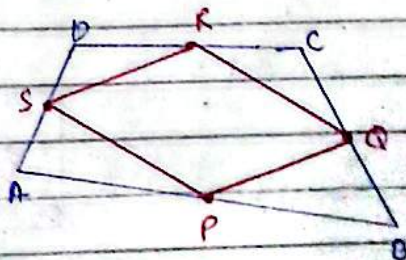
(i)



$\vec{DE}$ parallel to $\vec{AB}$ & $|\vec{DE}| = \frac{1}{2} |\vec{AB}|$

$\vec{r}$: position vector of point of division.

(ii)



PQRS is a parallelogram.

By joining the mid-points of any quadrilateral we get a parallelogram.

Parallel vectors:

if $\vec{a} = k\vec{b}$
 $\Rightarrow \vec{a}$ is parallel to $\vec{b}$

(where k is scalar)

(non-zero real numbers)

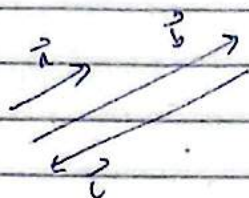
if a, b is non-zero

it has two possibilities

either positive

either negative

$\vec{a}$ & $\vec{b}$ are like parallel
 $\vec{a}$ & $\vec{c}$ are unlike parallel
 or anti-parallel



Also, if $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

then $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} = k$

$a_1 = kb_1, a_2 = kb_2, a_3 = kb_3$

$\frac{a_1}{b_1} = k, \frac{a_2}{b_2} = k, \frac{a_3}{b_3} = k$

Q#61 Find a vector of magnitude "5" & parallel to a vector $\vec{u} = 2\hat{i} + 3\hat{j} + 5\hat{k}$

Required vector = $5\hat{u}$ (like)

also, another vector = $-5\hat{u}$ (unlike)

i) Direction ratios

ii) Direction cosines

iii) Direction angles

if $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

Direction ratios: components of a vector are called direction ratios.

i.e. x, y & z

Direction cosines: components of a unit vector of $\vec{r}$ are called direction cosines.

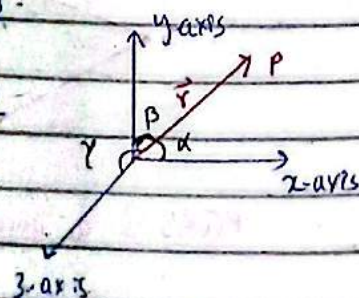
i.e. $\hat{r} = \frac{\vec{r}}{|\vec{r}|} = \frac{x\hat{i} + y\hat{j} + z\hat{k}}{\sqrt{x^2 + y^2 + z^2}}$

$\hat{r} = \frac{x}{r}\hat{i} + \frac{y}{r}\hat{j} + \frac{z}{r}\hat{k}$

$\hat{r} = l\hat{i} + m\hat{j} + n\hat{k}$

So, l, m & n are called direction cosines of $\vec{r}$.

Direction angles: α, β & γ are the direction angles of vector $\vec{r}$.



Also,
$$\left. \begin{aligned} \cos \alpha &= l \\ \cos \beta &= m \\ \cos \gamma &= n \end{aligned} \right\}$$

Note:

(i) $|\hat{r}| = \sqrt{l^2 + m^2 + n^2} = 1$
 $l^2 + m^2 + n^2 = 1$

(ii) $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$ (criteria to check the direction angles)

(iii) $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2$

Q2 Find the angle between the vector $\vec{r} = \underline{i} + \underline{j}$ and the x-axis.

$$\hat{r} = \frac{\underline{i} + \underline{j}}{\sqrt{1+1}} = \frac{1}{\sqrt{2}} \underline{i} + \frac{1}{\sqrt{2}} \underline{j}$$

$$\cos \alpha = \frac{1}{\sqrt{2}}$$

$$\boxed{\alpha = 45^\circ}$$

Note:

minimum number of angles to locate a point in space is 2, in plane is 1 & on line, no angle is required.

Dot product / Scalar Product:

$$\vec{a} \cdot \vec{b} = ab \cos \theta \quad ; \quad 0 \leq \theta \leq \pi$$

if $\vec{a} = a_1 \underline{i} + a_2 \underline{j} + a_3 \underline{k}$ & $\vec{b} = b_1 \underline{i} + b_2 \underline{j} + b_3 \underline{k}$

then $\vec{a} \cdot \vec{b} = (a_1 b_1) + (a_2 b_2) + (a_3 b_3)$

Properties:

(i) commutative property $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$ whether you move clockwise or anticlockwise as absorbs negative number

(ii) $\vec{a} \cdot \vec{b} = 0 \Rightarrow \theta = 90^\circ \Rightarrow \vec{a} \perp \vec{b}$

(iii) $\underline{i} \cdot \underline{j} = 0, \underline{j} \cdot \underline{k} = 0, \underline{k} \cdot \underline{i} = 0$

(iv) $\vec{a} \cdot \vec{a} = a \cos(0) = a^2$ also $(\vec{a})^2 = a^2$
 Vector Square = magnitude square.

(vi) $\underline{i} \cdot \underline{i} = 1, \underline{j} \cdot \underline{j} = 1, \underline{k} \cdot \underline{k} = 1$

(vi) As $\underline{a} \cdot \underline{b} = ab \cos \theta \Rightarrow \cos \theta = \frac{\underline{a} \cdot \underline{b}}{ab}$ i.e. (cosine of angle)

$\Rightarrow \theta = \cos^{-1} \left(\frac{\underline{a} \cdot \underline{b}}{ab} \right)$

(vii) projection of $\underline{a}$ along $\underline{b}$

(OR)

component of $\underline{a}$ along $\underline{b} = \frac{\underline{a} \cdot \underline{b}}{|\underline{b}|} = \underline{a} \cdot \hat{\underline{b}}$

Similarly,

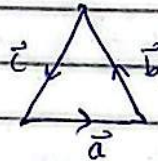
$\underline{a} \cdot \hat{\underline{i}} = \text{component of } \underline{a} \text{ along the } x\text{-axis.}$

Note: if $\underline{u} \cdot \underline{i} = 0, \underline{u} \cdot \underline{j} = 0, \underline{u} \cdot \underline{k} = 0$

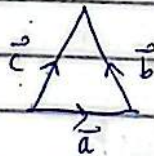
So, $\underline{u}$ is a null-vector.

Note: if $\underline{a}, \underline{b}$ & $\underline{c}$ are along the sides of a $\triangle ABC$ then

case (i) $\underline{a} + \underline{b} + \underline{c} = 0$



case (ii) $\underline{a} + \underline{b} = \underline{c}$



& if dot product of any two sides = 0 $\Rightarrow$ triangle will be a right angle triangle.

Note: (i) Angle in a semi-circle is right angle.

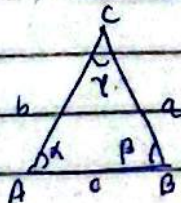
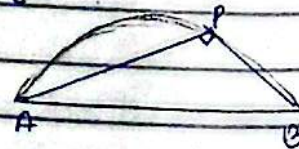
(ii) In any $\triangle ABC$,

$a = b \cos \gamma + c \cos \beta$

$b = c \cos \alpha + a \cos \gamma$

$c = a \cos \beta + b \cos \alpha$

Projection laws.



Cross product / vector product :

$$\vec{a} \times \vec{b} = ab \sin \theta \hat{n}$$

$$0 \leq \theta \leq 180^\circ$$

- (i) where $\hat{n}$ is a unit vector normal to the plane containing $\vec{a}$ & $\vec{b}$

↓
perpendicular

In one dimension $\Rightarrow$ we use word "perpendicular".

when we take about plane or space & say this is $\perp$ to surface / plane

↓
we use "normal" word

for it

- (ii) Direction of $\hat{n}$ can be obtained with the help of right hand rule

Properties :

(i) $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$ But $|\vec{a} \times \vec{b}| = |\vec{b} \times \vec{a}|$

(ii) $\vec{a} \times \vec{b} = \vec{0}$

$\Rightarrow \theta = 0^\circ$ or $\theta = 180^\circ$, $\Rightarrow \vec{a}$ is parallel to $\vec{b}$.

like parallel unlike parallel

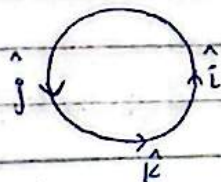
(iii) $\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = \vec{0}$

In general $\vec{a} \times \vec{a} = \vec{0}$

(iv) $\hat{i} \times \hat{j} = \hat{k}$

$\hat{j} \times \hat{k} = \hat{i}$

$\hat{k} \times \hat{i} = \hat{j}$



- (v) $\vec{a} \times \vec{b}$ is a vector which is $\perp$ to both $\vec{a}$ & $\vec{b}$

(vi) if $\vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$ & $\vec{b} = b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}$

then $\vec{a} \times \vec{b} =$

$\hat{i}$	$\hat{j}$	$\hat{k}$
a_1	a_2	a_3
b_1	b_2	b_3

mod

|| Sign comes

only on

either positive or scalar

no.

(vii) As $\vec{a} \times \vec{b} = ab \sin \theta \hat{n} \Rightarrow |\vec{a} \times \vec{b}| = ab \sin \theta |\hat{n}|$

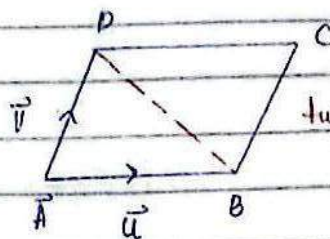
a) $\hat{n} = \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$

b) $\sin \theta = \frac{|\vec{a} \times \vec{b}|}{ab}$ i.e. (Sine of angle θ)

Application of cross Product:

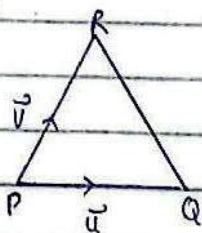
(i) Area of parallelogram

$$ABCD = |\vec{u} \times \vec{v}|$$



two concurrent triangles

(ii) Area of $\Delta PQR = \frac{1}{2} |\vec{u} \times \vec{v}|$



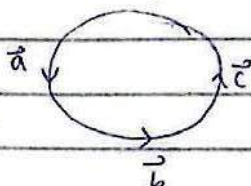
make vectors along any two adjacent sides of it.

Results: (i) if $\vec{a} \times \vec{b} \times \vec{c} = 0$

$$\text{then } \vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$$

cyclic

(ii) $\vec{a} \times (\vec{b} + \vec{c}) + \vec{b} \times (\vec{c} + \vec{a}) + \vec{c} \times (\vec{a} + \vec{b}) = 0$



(i) Vector triple product: $\vec{u} \times (\vec{v} \times \vec{w}) \Rightarrow \text{vector}$

(ii) Scalar triple product: $\vec{u} \cdot (\vec{v} \times \vec{w}) \Rightarrow \text{Scalar}$

↓

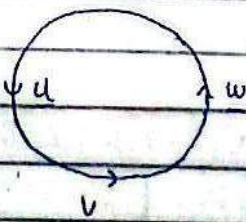
$$\vec{u} \cdot (\vec{v} \times \vec{w}) = [\vec{u} \ \vec{v} \ \vec{w}] = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}$$

Properties:

(i) cyclic property.

$$\underline{u} \cdot (\underline{v} \times \underline{w}) = \underline{v} \cdot (\underline{w} \times \underline{u}) = \underline{w} \cdot (\underline{u} \times \underline{v})$$

clockwise will give negative



Answers:

$$\begin{aligned} \text{(ii)} \quad & \underline{u} \cdot (\underline{v} \times \underline{w}) + \underline{v} \cdot (\underline{w} \times \underline{u}) + \underline{w} \cdot (\underline{u} \times \underline{v}) \\ & = 3 \underline{u} \cdot (\underline{v} \times \underline{w}) \end{aligned}$$

(iii) $\underline{u} \cdot (\underline{v} \times \underline{w}) = 0 \Rightarrow \underline{u}, \underline{v} \text{ \& \; } \underline{w} \text{ are coplanar Vectors}$

↓
Ask hi plane me a gye hain.

(iv) volume of Parallelepiped.
 $= \underline{u} \cdot (\underline{v} \times \underline{w})$

(v) volume of Tetrahedron
 $= \frac{1}{6} \underline{u} \cdot (\underline{v} \times \underline{w})$

(vi) $\underline{u} \cdot (\underline{v} \times \underline{w}) = 0$ if $\underline{u} = \underline{v}$ (or) $\underline{v} = \underline{w}$ (or) $\underline{u} = \underline{w}$

work done $= \underline{F} \cdot \underline{d}$

Moment $= \underline{r} \times \underline{F}$

↓

turning effect
of force.

↪ moment arm.

↪ $\underline{r}$ vector from Fixed point
to point of application of force.

Note: if two vectors are related like:

$\underline{B} = k\underline{A}$ then k tells the relationship between them.

case # (i) $k > 0$

↪ Same direction

↪ vectors are parallel & coincident.

} parallel & coincident

case # (ii) $k < 0$

↪ opposite direction.

↪ vectors are parallel but not coincident.

} parallel, opposite direction
but not coincident